AI-DRIVEN DATA ENGINEERING

REVIEW – 2

AI-Assisted Data Quality and Anomaly Detection on Hadoop-Based Healthcare Operational Data

Team No- 4

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Introduction

- Modern healthcare organizations generate large volumes of operational data through hospital IT systems such as appointment management, server logs, and monitoring systems.
- Maintaining high data quality is essential for accurate analysis, system reliability, and effective decision-making.
- Traditional rule-based data validation methods become inefficient and difficult to manage when dealing with large-scale and complex datasets.
- Big Data technologies like Hadoop enable scalable storage and distributed processing of high-volume healthcare operational data.
- AI-based anomaly detection techniques help automatically identify inconsistencies, missing values, and abnormal system behavior.
- This project proposes an AI-assisted framework using Hadoop and machine learning to improve data quality and detect anomalies efficiently.

Objectives

- To design and implement a Hadoop-based framework for storing and processing large-scale healthcare operational data in a distributed environment.
- To assess and improve data quality by identifying missing values, inconsistencies, and invalid records within the dataset.
- To apply AI-based anomaly detection techniques (e.g., Isolation Forest) to detect abnormal system behavior in healthcare operational data.
- To enhance data reliability and trustworthiness, enabling more accurate analysis and better operational decision-making.

Literature Survey

Sl No	Year(from 2020 onwards papers)	Authors	Title of the paper	Source(Journal/Conference name)	Objectve of the paper	Techniques Used	Dataset Used	Parameters Analysed	Research Inference	Research Gap
1	2021	Maria Mohammad Yousef	Big Data in Healthcare: Management, Analysis and Future Prospects	Journal of Big Data (Springer)	To review big data technologies used in healthcare including Hadoop ecosystem	HDFS, MapReduce, Big Data Analytics	Healthcare & EHR datasets	Patient records, disease trends, hospital resource usage	Hadoop improves scalability and distributed healthcare analytics	Lacks academic single-node implementation and controlled anomaly testing
2	2020	Micheal James	Network Intrusion Detection System: A Systematic Study of Machine Learning and Deep Learning Approaches	Transactions on Emerging Telecommunications Technologies	To analyze anomaly detection techniques using ML algorithms including Isolation Forest	Isolation Forest, ML Classification	Large structured datasets	Outliers, abnormal patterns	Isolation Forest is efficient for anomaly detection in large datasets	Does not integrate Hadoop-based distributed storage framework

Proposed Method

DATASET
- Synthetic healthcare operational data
- CPU, errors, response

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HDFS
- Distributed storage
- Scalable & fault tolerant system

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APACHE SPARK
- Fast data processing
- Spark SQL queries
- In-memory computation

CLEANED DATA
- Removed NULL values
- Removed invalid data
- Structured dataset

→

PYTHON AI
- Isolation Forest model
- Detect abnormal system behavior

→

GRAPH OUTPUT
- Visualize anomalies
- Normal vs abnormal
- Decision support

Implementation

- **Tools & Software:** The project uses **Apache Hadoop** for distributed storage and processing, running on **Ubuntu** with **Java**. Anomaly detection is implemented using **Scikit-learn** in Python.
- **Environment Setup:** Hadoop is configured in a single-node VM with proper HDFS and YARN setup. Required configuration files are edited and services are successfully started to create a working big data environment.
- **Dataset Selection:** A synthetic hospital dataset is generated to avoid privacy issues and allow controlled anomaly (3%) and dirty data (2%) injection. The dataset size (1100–3500 rows) suits a 4GB VM.
- **Sample Dataset:** The dataset is in CSV format with fields like Hospital_ID, Date, Patient_Count, Resource_Usage, and Emergency_Cases, enabling structured big data processing.
- **Implementation Process:** Data is uploaded to HDFS, cleaned, processed using Hadoop, and analyzed using Isolation Forest. Results are stored and visualized to demonstrate anomaly detection in a complete pipeline.

Conclusion

- **Project Summary:** This project presents a Hadoop-based big data pipeline for hospital monitoring using a synthetic dataset, demonstrating distributed storage, processing, and AI-based anomaly detection in a structured workflow.
- **Implementation Outcome:** The system successfully uses **Apache Hadoop** to store data in HDFS, perform data cleaning, and process records in a distributed environment.
- **Scalability & Significance:** Although implemented on a small VM dataset, the architecture is scalable and capable of handling large-scale real-world healthcare data.
- **Next Steps:** Future work includes increasing dataset size, improving data processing logic, adding visualization dashboards, and conducting performance evaluation in the next phase.

References

- https://www.researchgate.net/publication/398320051_Anomaly_Detection_in_Healthcare_Data_Using_Machine_Learning?
- https://aircconline.com/ijcsit/V13N2/13221ijcsit02.pdf?

THANK YOU